

James R. Rose

COMPUTATIONAL BIOLOGIST, PHD

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RESEARCH STATEMENT

Computational biologist developing methods for image-based spatial transcriptomics, with a background in the transcriptional and epigenetic regulation of lymphocytes. My current work asks how far we can trust the cell boundaries that every downstream spatial result rests on, and how to make agent-run analysis reproducible.

EDUCATION

PhD, Genetics & Molecular Biology

2019 – 2024

Emory University

Atlanta, GA

Advisor: Jeremy M. Boss. Defended August 2024.

THESIS Epigenetic gene regulation in B and T lymphocytes and its role in cellular activation, differentiation, and autoimmune disease

MS, Biotechnology

2011 – 2013

The Pennsylvania State University

University Park, PA

Integrated masters with BS degree

THESIS Modeling interfacial shear stress on mammalian cell culture systems

BS, Biotechnology

2008 – 2013

The Pennsylvania State University

University Park, PA

RESEARCH & PROFESSIONAL EXPERIENCE

Assistant Scientist

Sep 2024 – Present

Santangelo Lab, Department of Biomedical Engineering, Emory University

Atlanta, GA

Sole computational scientist for a ~25-person molecular engineering lab developing mRNA and nucleic-acid therapeutics. I own the spatial transcriptomics platform end to end, build the lab's analysis software and infrastructure, and report to program sponsors including ARPA-H, DARPA, and DTRA.

- Developed **CRISP**, an open-source R and Python package that quantifies cell segmentation quality in image-based spatial transcriptomics without ground-truth annotation, by detecting cells that co-express mutually exclusive marker genes.
- Benchmarked six segmentation approaches (10x multimodal segmentation kit, Baysor, Proseg, Segger, Cellpose, nuclear expansion) across ten mouse organs on a 5,006-gene Xenium panel. How much transcript capture you give up for cell purity turns out to depend on the tissue.
- Built a multi-tissue spatial atlas of lipid nanoparticle delivery comparing injection routes across **2.8 million cells, 11 organs, and 104 tissue-microarray punches**. The two routes trigger different immune responses and remodel splenic myeloid populations differently.

- Built the lab's Xenium toolchain, from raw 10x output to annotated cells: a Nextflow pipeline for alternative segmentation following nf-core conventions, Seurat workflows for QC through annotation, and a Python tool that splits multi-punch tissue microarrays into per-sample datasets.
- Moved terabyte-scale segmentation onto 4 TB-memory AWS instances, for runs no single workstation can hold. Chunking the work brought a 332-million-transcript tissue array down to a 72-hour run.
- Map mRNA delivery tropism across tissues and species, including a custom method that measures immunofluorescence intensity inside Xenium cell boundaries to tie transcript delivery to protein production. This is the spatial analysis behind a polymer lung-delivery manuscript under review at *Advanced Materials*.
- Designed the lab's custom Xenium Prime add-on probe panel for mouse airway, selecting 44 markers from a 203-candidate review of the published airway literature.
- Lead guide design for a CRISPR-activation therapeutic program. I scored and ranked a 552-guide activity screen, then checked the candidates for cross-species conservation and against gnomAD and *All of Us* population variants.
- Created **Stringency**, a safety system for AI-driven data analysis: it validates the analysis plan against best practice before any code runs, keeps a tamper-proof provenance log, and pauses at key decisions until a person signs off.
- Run the lab's AI tooling. I procured and administer the team's agent-assistant deployment, and wrote the skills that let bench scientists run validated pipelines on their own data and look up lab protocols without waiting on me.
- Automated the group's NGS analysis as reproducible Nextflow workflows, covering bulk and single-cell RNA-seq, NanoString, and targeted amplicon panels for a dozen collaborators.
- Administer the lab's compute: two analysis workstations with 134 cores and 180+ TB of storage, shared network mounts, and automated backups.
- Own the quarterly technical reporting and present results to program sponsors and collaborating labs.
- Mentor junior lab members in R, Python, and reproducible analysis, and teach the group to use new computational tools.

PhD Graduate Researcher

Aug 2019 – Aug 2024

Boss Lab, Genetics & Molecular Biology Program, Emory University

Atlanta, GA

Thesis work on how transcriptional and epigenetic state encodes immune cell identity, and how that regulation breaks in autoimmunity.

- Integrated RNA-seq and ATAC-seq across human memory T cell subsets to identify *poised accessible regions*, chromatin whose accessibility before and after activation separates the subsets and predicts how they respond. Published in *Communications Biology*.
- Showed that lupus disease activity leaves a signature in both the transcriptome and the epigenome of otherwise resting naïve B cells, in a collaboration with Emory rheumatology. Accepted at the *Journal of Clinical Investigation*.
- Compiled an SLE single-cell B cell reference atlas from public datasets, then queried in-house samples against it and classified disease endotypes with machine learning.
- Designed and maintained the lab's bioinformatic pipelines for next-generation sequencing: QC, alignment, peak calling, differential testing, and gene regulatory network analysis.

Senior Associate Scientist, Biotherapeutics Development

Aug 2016 – Jun 2019

Janssen R&D, Johnson & Johnson

Malvern, PA

- Led a team of development researchers on monoclonal antibody manufacturing process development, new technology implementation, and regulatory submissions for a therapeutic product worth \$6B in annual sales.
- Implemented statistical process control across international GMP manufacturing facilities.

— Authored sections of the BLA and other regulatory filings.

Engineer I, Cell Culture Technical Development

Jun 2013 – Aug 2016

Biogen

Durham, NC

- Developed mammalian cell culture processes for monoclonal antibody production at large-scale manufacturing.
- Developed process analytics technology, including Raman spectroscopy for remote metabolite monitoring.
- Pioneered the group's use of stochastic (Monte Carlo) simulation modeling for manufacturing facility-fit analysis.

PUBLICATIONS

Rose, J.R., Rose, E.S., Assumpção, J.A.F., Pathak, H., Peck, H.E., Sasser, L.E., Patel, C.J., Vanover, D. & Santangelo, P.J. (2026) CRISP enables comparisons of image-based spatial transcriptomic segmentation quality across ten organs. *bioRxiv*2026.04.16.718944 **PREPRINT**

Rose, J.R., Ishii, Y., Hicks, S.L., Scharer, C.D., Jenks, S.A., Sanz, I. & Boss, J.M. (2026) Systemic lupus erythematosus signatures in resting naïve B cells differ by disease activity. *Journal of Clinical Investigation* **ACCEPTED**

Pramod Kumar, E.K., Jung, Y., Joo, J.Y., Rose, E.S., **Rose, J.R.**, et al. & Santangelo, P.J. (2026) An engineered thio-amino polymer achieves potent, safe and broad cell type delivery in the lung mapped by spatial transcriptomics. *Advanced Materials* **UNDER REVIEW**

Rose, J.R., Akdogan-Ozdilek, B., Rahmberg, A.R., Powell, M.D., Hicks, S.L., Scharer, C.D. & Boss, J.M. (2023) Distinct transcriptomic and epigenomic modalities underpin human memory T cell subsets and their activation potential. *Communications Biology*6, 363.

Patterson, D.G., Kania, A.K., Price, M.J., **Rose, J.R.**, Scharer, C.D. & Boss, J.M. (2021) An IRF4–MYC–mTORC1 integrated pathway controls cell growth and the proliferative capacity of activated B cells during B cell differentiation in vivo. *The Journal of Immunology*207(7), 1798–1811.

PUBLISHED ABSTRACTS & POSTERS

Rose, J.R., Ishii, Y., Hicks, S.L., Scharer, C.D., Sanz, I. & Boss, J.M. (2023) Early peripheral B cells in inactive SLE are transcriptionally and epigenetically primed for T-dependent activation. *The Journal of Immunology*210, 149.08.

Akdogan-Ozdilek, B., **Rose, J.R.**, Rahmberg, A.R., Scharer, C.D. & Boss, J.M. (2023) Human memory T cells show differential metabolic preferences. *The Journal of Immunology*210, 87.05.

Wiggins, K.J., **Rose, J.R.**, Boss, J.M. & Scharer, C.D. (2023) Class switched memory B cell development and transcriptional programming is regulated by EZH2. *The Journal of Immunology*210, 148.06.

Powell, M.D., **Rose, J.R.**, Gupta, M., Rahmberg, A.R., Scharer, C.D. & Boss, J.M. (2023) Tissue-specific heterogeneity of human T follicular helper cells marked by unique transcriptomes and regulatory architecture. *The Journal of Immunology*210, 148.22.

Rose, J.R., Rahmberg, A.R., Powell, M.D., Scharer, C.D. & Boss, J.M. (2022) Epigenome accessibility changes before and after activation reveal distinct and progressive differentiation for human memory T cell subsets. *The Journal of Immunology*208, 57.02.

Powell, M.D., **Rose, J.R.**, Neeld, D.K., Rahmberg, A.R., Scharer, C.D. & Boss, J.M. (2022) Defining the cis- and trans-regulatory architecture for human PD-1 in T follicular helper cells. *The Journal of Immunology*208, 112.21.

Ishii, Y., **Rose, J.R.**, Cashman, K., Hicks, S.L., Deguchi, T., Scharer, C.D., Jenks, S.A., Boss, J.M. & Sanz, I. (2022) Distinct transcriptomic and epigenetic profiles mediated by disease activity in SLE. *Arthritis & Rheumatology*74, 22–23.

Rose, J.R., Rahmberg, A.R., Scharer, C.D. & Boss, J.M. (2021) Unique abilities of human memory T cell subsets are associated with both specialized gene expression modules and epigenetic priming. *The Journal of Immunology*206, 98.28.

Powell, M.D., Neeld, D.K., **Rose, J.R.**, Scharer, C.D. & Boss, J.M. (2021) Exploring mechanisms of human PD-1 gene regulation in T follicular helper cells. *The Journal of Immunology* 206, 53.06.

INVITED TALKS & ORAL PRESENTATIONS

Spatial transcriptomics quality control for a sepsis model study. *Invited seminar, collaborating research group, Emory University.* September 2026

Hands-on workshop: AI coding agents for laboratory data analysis. *Santangelo Lab workshop, Emory University.* February 2026

Spatial atlas of nucleic acid delivery: program technical reviews. *DARPA and ARPA-H program reviews.* 2025 – 2026

Integrated spatial transcriptomics and imaging reveal cellular targets of mRNA lung delivery. *Georgia Tech research symposium, Atlanta, GA.* April 2025

Setting and re-setting chromatin accessibility in inactive SLE. *Emory Chromatin Club, Atlanta, GA.* September 2023

Early peripheral B cells in inactive SLE are transcriptionally and epigenetically primed for T-dependent activation. *American Association of Immunologists Annual Meeting, Washington, DC.* May 2023

Epigenome accessibility changes before and after activation reveal distinct and progressive differentiation for human memory T cell subsets. *American Association of Immunologists Annual Meeting, Portland, OR.* May 2022

SOFTWARE

CRISP

R · Python

Segmentation quality assessment for spatial transcriptomics via negative-coexpression purity metrics. Supports Xenium, Seurat, and SpatialExperiment objects.

Stringency

Python

A safety harness for running AI agent data analysis pipelines. Validates the analysis plan before any code runs, keeps a tamper-proof provenance log, and pauses at key decisions until a person signs off.

Xenium analysis toolkit

Nextflow · R · Python

Three pipelines covering raw Xenium output to annotated cells: alternative segmentation (Xen_Segmentation_NextFlow), tissue microarray splitting (Xen_TMA_pipeline), and Seurat-based QC through annotation (Xen_Seurat_Pipeline).

CellQuant

R

Cell-composition modelling for spatial transcriptomics. Mixed-effects and spatial (SPDE) models with niche-aware adjustment, for deciding whether a cell type actually shifted between conditions.

pyMFI

Python

Measures immunofluorescence intensity inside spatial-transcriptomics cell segmentation boundaries, so protein signal and transcript profile can be read off the same cell.

TEACHING

Course Lecturer & Teaching Assistant

Fall 2022, Fall 2023

IBS 519: Statistical Rigor, Reproducibility, and Experimental Design

Emory University

- Developed teaching materials for a revamped curriculum on quantitative data analysis.
- Lectured on data visualization principles and ggplot2 for incoming graduate students.

Teaching Assistant

IBS 500R: Introduction to Data Analysis Using R

— Taught introductory R programming to incoming biomedical graduate students.

Fall 2020, Fall 2021

Emory University

AWARDS & HONORS

Abstract Award. *American Association of Immunologists Annual Meeting* 2022, 2023

AI Innovation Fellowship. *Inspirit Innovators* 2023

Undergraduate Research Award. *Eberly College of Science, Penn State* 2013

Phi Beta Kappa. *National Honors Society* 2013

SERVICE & LEADERSHIP

President & Founding Board Member

2025–present

Friends of Mulberry Fields, Atlanta GA

Founded and incorporated a 501(c)(3) that stewards a 39-plot community garden and greenspace. Drafted the bylaws, negotiated the land management agreement with the land conservancy, secured the organization's insurance and tax-exempt status, and chair the governance committee.

Peer reviewer

2026

Scientific manuscript review

Research mentor

2025–present

Santangelo Lab, Emory University

Computational training for junior lab members

Co-organizer & instructor

2023

Peer bioinformatics teaching series, Emory University

Sessions on advanced ggplot2, control flow, and dimensionality reduction

Student Representative

2021–2023

Genetics & Molecular Biology Graduate Program, Emory University

TECHNICAL SKILLS

Languages & tools	R (tidyverse) · Python · Bash · Nextflow/Groovy · SQL · Git · Docker · Quarto / R Markdown · Jupyter · Shiny
Spatial & imaging	10x Xenium · Cell segmentation (Baysor · Proseg · Segger) · Spatial statistics · Tissue microarrays · Immunofluorescence integration · QuPath · ImageJ · OpenCV
Genomics	Single-cell RNA-seq · Bulk RNA-seq · ATAC-seq · CITE-seq · Seurat / Bioconductor / scverse · Differential expression (DESeq2) · Data integration · Gene regulatory networks · CRISPR guide design
Statistics & ML	Statistical modeling and inference · Machine learning · Experimental design · Monte Carlo simulation · Benchmarking · Data visualization (ggplot2 · seaborn)
AI for bioinformatics	Agentic analysis workflows · LLM tool and skill development · Reproducibility and audit frameworks · Evaluation of agent output
Infrastructure	AWS · Multi-GPU workstation administration · RStudio Server · Workflow automation

Domain

Immunology (B and T cell) · Epigenetics and gene regulation · Autoimmunity · mRNA and nucleic acid delivery · Translational and clinical data · GMP biologics development